TED(10)- 4045
Revision 2010

MODEL QUESTION PAPER
COMMUNICATION ENGINEERING

(Maximum marks : 100) Time : 3 Hours

PART – A

I. Answer the following questions in one or two sentences. Each question carries 2 marks.

1. List the factors affecting field strength of ground waves.
2. What is a resonant antenna?
3. What is the function of balanced modulator?
4. How FM is generated in direct method?
5. Define sensitivity of a receiver. (5 X 2 = 10)

PART - B

II. Answer any five of the following. Each question carries 6 marks.

1. (i). What is skip distance?
   (ii). Give the reason for better HF reception at night.
2. With a neat sketch explain rhombic antenna.
3. Derive the expression for AM wave.
4. With the help of band spectrum describe vestigial side band system.
5. (i). What is the function of harmonic generator in AM transmitter?
   (ii). A 1 KW carrier is modulated with a depth of modulation of 0.8. Find the total radiated power.
6. Why we need pre-emphasis and de-emphasis in FM?
7. Describe the operation of a simple diode detector, showing the dependence of values of R and C with carrier and modulating frequencies. (5 X 6 = 30)

PART – C

(Answer one full question from each unit. Each question carries 15 marks)

UNIT – I

III. (a) Differentiate between transverse and longitudinal waves. 3

(b) Explain the different layers of Ionosphere ,with the roles of each in communication. 12

OR

IV. (a) How the characteristics of a single radiating element is modified by the introduction of additional elements. 4

(b) Explain broad side array and end fire array with the help of their radiation patterns. 8

(c) What is meant by the term Diversity. 3

UNIT – II

V. (a) With the waveforms compare FM and PM. 8

(b) With the help of block diagram differentiate between low level and High level modulators. 7

OR

VI. Why we need suppressed carrier system .With circuit diagram and Waveforms explain the suppression of carrier. 15

UNIT – III

VII. (a) Explain the three types of internal noise encountered in communication system. 9

(b) Draw the block diagram of AM transmitter ,mentioning the need of modulating and modulated amplifiers. 6

OR

VIII. (a) With the help of block diagram show how the frequency is stabilized in indirect method of FM generation. 12

(b) Where we need AFC? Why? 3

UNIT – IV

IX. (a) What is AGC? Why it is given at the initial stages of amplifiers? 5

(b) With the help of block diagram explain FM receiver. 10

OR

X. (a) Differentiate between forward and reverse AGC. 3

(b) Explain various AGC characteristics with the help of graph. 6

(c) List the advantages of RF amplifier. 6